

DuoShield – Full Project Context Handoff

Last Updated: 2026-06-11 | Firebase Project: duoshield-8caf1 | Package: com.duoshield.app | Total Java files: 53

INSTRUCTIONS: Copy-paste this entire document as the FIRST message in a new Replit Agent session.

What this project is

DuoShield is a secure, end-to-end encrypted Android messaging app (Java, minSdk 26, targetSdk 34). It is a 2-person (duet) messaging app — one user pairs with exactly one partner. It uses Firebase Blaze plan (free-tier limits must be respected at all times). All code is Java/XML written in Replit and manually copied into Android Studio by the user. There is also a web landing page built in React/Vite (see artifacts/duoshield in Replit).

Stack

Layer	Technology
Language	Java (Android)
Min SDK	26 (Android 8)
Target SDK	34
UI	Material Design 3
Database (local)	Room DB v7 [next change = v8 + MIGRATION_7_8]
Database (remote)	Firebase Firestore
Auth	Firebase Auth (email/password)
Storage	Firebase Storage
Push	Firebase Cloud Messaging (FCM)
Images	Glide 4.16 + PhotoView (JitPack)
Background	WorkManager 2.9
Biometric	AndroidX Biometric 1.2.0-alpha05
Package	com.duoshield.app
Version	1.0.0 (versionCode 1)
Firebase Project	duoshield-8caf1

Gradle Dependencies

See duoshield_ui/app_dependencies.gradle for the full list. Key dependencies:

- com.google.android.material:material:1.12.0
- androidx.room:room-runtime:2.6.1
- androidx.work:work-runtime:2.9.0
- androidx.biometric:biometric:1.2.0-alpha05
- Firebase BOM 33.1.0 (auth, firestore, storage, messaging)
- com.github.bumptech.glide:glide:4.16.0
- com.github.chrisbanes:PhotoView:2.3.0 (JitPack — for MediaViewerActivity)
- androidx.security:security-crypto:1.1.0-alpha06

Complete File Map (55+ files in duoshield_ui/)

All files live in duoshield_ui/. Copy them into your Android Studio project at matching paths.

Java – Activities & Application (~19 files)

```
app/src/main/java/com/duoshield/app/
■■■ DuoShieldApp.java ← Application subclass (channels, WorkManager, lock)
■■■ BaseActivity.java ← Lock check in onResume (extends AppCompatActivity)
■■■ SignInActivity.java ← LAUNCHER - Firebase email/password sign-in
■■■ PairingActivity.java ← First-run: enter partner UID to link conversation
■■■ KeyFingerprintActivity.java ← View + verify ECDH key fingerprint (SHA-256)
```

- ConversationListActivity.java ← Main chat list (Firestore + DiffUtil + swipe)
- LockScreenActivity.java ← Biometric + PIN lock (shown after 3min bg)
- MessageSearchActivity.java ← Offline full-text search (Room, highlighted)
- SettingsActivity.java ← Security, notifications, quota, wipe
- MediaViewerActivity.java ← Full-screen pinch-to-zoom image viewer
- FullScreenImageActivity.java ← [NEW] Full-screen image viewer (ScaleGestureDetector)
- ChatMediaActivity.java ← [COMPLETE] Full chat activity, all helpers wired
- ChatMediaActivity_IntegrationSnippet.java ← Integration guide
- DuoShieldMessagingService.java ← FCM service (calls NotificationStyler)
- MarkReadReceiver.java ← Notification 'Mark as read' BroadcastReceiver
- MessageReplyReceiver.java ← Notification inline-reply BroadcastReceiver

Java – UI Layer (~15 files)

app/src/main/java/com/duoshield/app/ui/

- MessageAdapter.java ← Chat RecyclerView (DiffUtil, Glide, 4 view types, status ticks, edited label, multi-select)
- ConversationAdapter.java ← Conversation list (DiffUtil, typing, online, badges)
- MessageActionsBottomSheet.java ← Long-press sheet (react/reply/copy/pin/edit/forward)
- DisappearingTimerSheet.java ← Timer picker (Off/1min/5min/1hr/1day)
- SwipeToDeleteCallback.java ← Left-swipe delete for conversation list
- SwipeToReplyCallback.java ← Right-swipe reply gesture in chat
- TypingDotsAnimator.java ← Staggered 3-dot wave animation
- TypingIndicatorView.java ← ViewGroup that wraps TypingDotsAnimator
- SendButtonAnimator.java ← Mic<->send morph on text input
- ChatUiHelper.java ← Smooth scroll, reply bar, FAB toggle
- VoiceWaveformView.java ← Custom bar waveform (recording + playback)
- SelfDestructCountdownView.java ← Circular countdown arc on disappearing messages
- SearchResultsAdapter.java ← Search results RecyclerView adapter

Java – ViewModels (2 files)

app/src/main/java/com/duoshield/app/viewmodel/

- ConversationViewModel.java ← LiveData for conversation list (Firestore listener)
- MessageViewModel.java ← LiveData for message list (Firestore listener)

Java – Utilities (~52 files)

app/src/main/java/com/duoshield/app/util/

- FirebaseCostGuard.java ← Daily read/write/delete quota tracker (SharedPreferences)
- FirebaseQuotaSummary.java ← Human-readable usage labels for SettingsActivity
- PresenceThrottle.java ← Debounced typing/online writes (2-3s)
- OnlinePresenceHelper.java ← Sets online_{uid}=true onResume, clears onPause
- NotificationStyler.java ← FCM rich notifs (MessagingStyle+BigTextStyle, inline reply)
- BiometricHelper.java ← BiometricPrompt wrapper (finger+face+credential)
- AppLockManager.java ← Decides when to show lock screen (3-min timeout)
- SessionLockTimer.java ← Foreground inactivity countdown (resets on touch)
- PinManager.java ← PIN storage (EncryptedSharedPreferences) + verify
- MessagePaginator.java ← Backward pagination, 30 msgs/page (scroll-triggered)
- SelfDestructScheduler.java ← WorkManager jobs for disappearing message deletion
- StorageCleanupWorker.java ← Weekly WorkManager job: trim image cache >30 MB
- MediaPickerHelper.java ← Modern camera/gallery picker (ActivityResultContracts)
- UploadHelper.java ← Firebase Storage (images compressed, voice .3gp)
- VoiceRecorderHelper.java ← MediaRecorder wrapper (.3gp, amplitude samples)
- VoiceMessagePlayer.java ← MediaPlayer wrapper (play/pause/resume/stop)
- DeliveryReceiptHelper.java ← Batch mark-as-delivered / reset unread (WriteBatch)
- ReadReceiptHelper.java ← Mark messages as read in Firestore
- ForwardMessageHelper.java ← Forward message as new send
- EditMessageHelper.java ← Edit own message (48h window, sets edited=true)
- PinMessageHelper.java ← Pin/unpin messages (max 3, FIFO eviction)

```

■■■ ReactMessageHelper.java ← Add/toggle emoji reaction on message
■■■ ClipboardHelper.java ← Secure copy (sensitive flag, auto-clear 90s)
■■■ TimeFormatter.java ← Telegram-style timestamps (HH:mm / Mon / d MMM)
■■■ DateHeaderHelper.java ← Date chip logic (Today / Yesterday / weekday / date)
■■■ LastSeenFormatter.java ← 'last seen 3 minutes ago' / 'online' label
■■■ MessageStatusHelper.java ← Binds ic_done / ic_done_all with correct tint
■■■ MessageBuilder.java ← Fluent Firestore message doc builder (text/image/voice)
■■■ ConversationMetaUpdater.java ← Updates lastMessage + unread_{uid}++ after send
■■■ TextStyleHelper.java ← Inline markdown (*bold* _italic_ ~strike~ `mono`)
■■■ EmojiHelper.java ← Quick-reaction emoji set
■■■ EncryptionHelper.java ← AES-256 encrypt/decrypt (uses shared ECDH key)
■■■ ContactPairingHelper.java ← Creates Firestore conversation doc for partner UID
■■■ LinkPreviewHelper.java ← Extracts first URL from text (no remote fetch)
■■■ ImageCacheHelper.java ← Disk image cache (50 MB cap, LRU eviction)
■■■ GlideHelper.java ← Standardized Glide calls (avatar/media/toolbar)
■■■ HapticHelper.java ← Haptic patterns (long-press, send, wrong PIN, reaction)
■■■ KeyboardHelper.java ← Show/hide soft keyboard helpers
■■■ NetworkStateHelper.java ← Monitors connectivity (offline banner)
■■■ SecureShareHelper.java ← Share image via FileProvider (no signed URL leak)
■■■ MediaSizeEstimator.java ← Cumulative storage usage estimate (local)
■■■ AppUpdateHelper.java ← Read versionName/versionCode for Settings
■■■ MuteHelper.java ← Mute/unmute conversation (Firestore muted_{uid})
■■■ TypingThrottle.java ← Per-key typing write throttle (2s debounce)
■■■ UnreadCountHelper.java ← Read+reset unread_{uid} in Firestore
■■■ WipeHelper.java ← Full local data wipe (Room + SharedPrefs + cache)
■■■ FcmTokenHelper.java ← Register/unregister FCM token in Firestore
■■■ ExportHelper.java ← [NEW] Chat -> PDF (Android PdfDocument + FileProvider)
■■■ SearchHelper.java ← [NEW] Background Room full-text search + main-thread cb
■■■ SelfDestructWorker.java ← [UPDATED] dual-sweep: deleteExpired(now) + deleteOlderThan(cutoff)

```

Java – DB Helpers (2 files)

```

app/src/main/java/com/duoshield/app/db/
■■■ MessageDao_additions.java ← 15 @Query methods added to MessageDao
■ (insertAll, updateMessage, getMessagesPage, getMessageById,
■ getUndeliveredMessages, searchMessages, getLatestMessages,
■ markAllRead, updateStatus, deleteMessage, deleteExpired(now),
■ deleteOlderThan(cutoff), deleteAll)
■■■ AppDatabase_migration8.java ← MIGRATION_7_8 – paste into AppDatabase.java

```

Layouts (35 files)

```

app/src/main/res/layout/
activity_chat.xml activity_conversation_list.xml activity_settings.xml
activity_lock_screen.xml activity_message_search.xml activity_sign_in.xml
activity_pairing.xml activity_key_fingerprint.xml activity_media_viewer.xml
item_message_sent.xml item_message_received.xml item_image_message.xml
item_voice_message.xml item_conversation.xml item_date_header.xml
item_reaction_row.xml item_reaction_pill.xml item_link_preview.xml
item_emoji_chip.xml item_search_result.xml item_settings_toggle.xml
item_settings_nav.xml item_settings_danger.xml item_settings_info.xml
layout_typing_indicator.xml layout_message_input_bar.xml
layout_reply_preview_bar.xml layout_voice_recording_bar.xml
layout_pinned_message_banner.xml layout_fab_scroll.xml
layout_offline_banner.xml layout_empty_conversations.xml
layout_quota_stats.xml bottom_sheet_message_actions.xml
bottom_sheet_disappearing_timer.xml

```

Drawables (72 files), Values (7 files), Animations (17 files)

Same as previous handoff — no changes. See drawables/, values/, anim/, transition/.

Message.java Fields (Room Entity)

```
@Entity(tableName = "messages")
public class Message {
    @PrimaryKey public String id;
    public String conversationId;
    public String sender;
    public String text; // AES-256 encrypted, decrypted by EncryptionHelper
    public long timestamp;
    public boolean isEncrypted;
    public String mediaUrl; // Firebase Storage URL
    public boolean delivered;
    public long selfDestructAt; // 0 = no self-destruct
    public String replyToId;
    public String replyToPreview; // plaintext preview (max 60 chars)
    public String reaction; // e.g. "<3" - legacy single reaction
    public boolean edited; // added in MIGRATION_6_7
    public String status; // 'pending'|'sent'|'delivered'|'read' (MIGRATION_7)
}
```

MessageAdapter Constructor — MUST NOT CHANGE

```
new MessageAdapter(
    List messages,
    String myUid,
    OnVoicePlayListener voiceListener,
    OnMessageLongPressListener longPressListener
)
adapter.setMessages(newMessageList); // always use DiffUtil, never notifyDataSetChanged()
```

MessageAdapter — Status Tick Logic (IMPLEMENTED)

```
pending -> ic_done (single grey tick)
sent -> ic_done_all (double grey tick)
delivered -> ic_done_all (double grey tick, tint #90A4AE)
read -> ic_done_all (double teal tick, tint #2AABB8)
Backward-compatible with delivered/seen booleans.
(edited) label via null-safe R.id.editedLabel
Timestamp via TimeFormatter + null-safe R.id.messageTimestamp
TextStyleHelper.apply() for bold/italic/strike/mono spans on text messages.
```

Room DB Migration Rule

Current version: 7. Migration chain 1→2→3→4→5→6→7 all registered in AppDatabase.java.

Next change MUST be version 8 + MIGRATION_7_8. File AppDatabase_migration8.java contains the template.

Message.java changes so far: edited (boolean, v6) | status (String, v7)

Firestore Data Structure

```
conversations/{conversationId}
members: array [uid1, uid2]
lastMessage: string plain-text preview (max 80 chars)
lastMessageTs: timestamp
lastSenderId: string
pinnedMessages: array max 3 message IDs
disappearingTimer: number ms, 0 = off
typing_{uid}: boolean
online_{uid}: boolean
lastSeen_{uid}: timestamp
unread_{uid}: number
```

muted_{uid}: boolean
hidden_{uid}: boolean soft-delete for one side
partnerName_{uid}: string
avatarUrl_{uid}: string
subcollection messages/{messageId}:
sender: string
text: string (AES-256 encrypted)
timestamp: timestamp
type: 'text' 'image' 'voice'
mediaUrl / thumbnailUrl / mediaWidth / mediaHeight: ...
duration: number (voice, ms)
amplitudes: array (voice waveform samples)
status: 'pending' 'sent' 'delivered' 'read'
replyToId / replyToText: string
reactions: map> emoji -> [uid, ...]
reaction: string (legacy single reaction)
edited: boolean
destructAt: number (epoch ms, 0 = no destruct)
forwarded: boolean
users/{uid}
token: string (FCM device token)
platform: 'android'
updatedAt: timestamp

Firebase Free-Tier Rules (Blaze Plan)

Quota	Free limit	DuoShield safe cap
Firestore reads/day	50,000	40,000
Firestore writes/day	20,000	16,000
Firestore deletes/day	20,000	16,000
Storage stored	5 GB	< 1 GB
Storage download/day	1 GB	< 500 MB
FCM	Free	✓
Cloud Functions	2M/month	AVOID ENTIRELY

Enforced via `FirestoreCostGuard` — always call `guard.canRead(n)` / `guard.canWrite(n)` / `guard.canDelete(n)` before any Firestore operation. Show `FirestoreQuotaSummary.get(ctx)` in Settings.

UI Design System

Background	#0D0D0D
Surface	#1E1E1E
Sent bubble	#1A5C6E (dark teal)
Received bubble	#1E1E1E
Accent teal	#2AABB8
Text primary	FFFFFF
Text secondary	#9E9E9E
Timestamp	#90A4AE
Seen tick	#2AABB8 (teal)
Delivered tick	#90A4AE (grey)
Danger	#EF5350

Online	#4CAF50
Theme base	Theme.Material3.Dark.NoActionBar
Bubble corner	18dp (4dp on tail side)
Input corner	20dp

Auth Flow (Activity Stack)

```

SignInActivity (LAUNCHER)
|
+-- first-run (no conversationId in prefs)
|   PairingActivity
|   |
+-----+--> ConversationListActivity <-> ChatMediaActivity
|
+--> MediaViewerActivity / FullScreenImageActivity
+--> KeyFingerprintActivity
+--> LockScreenActivity (auto-lock after 3 min bg)
+--> SettingsActivity

```

Fully Implemented Features (DO NOT RE-IMPLEMENT)

Core / Security

- E2E encryption: ECDH key exchange + AES-256 via EncryptionHelper
- Firebase Auth pairing (email/password, partner UID stored in conversation)
- Biometric lock (BiometricHelper) + duress PIN wipe
- LockScreenActivity – shown after 3min background (AppLockManager)
- SessionLockTimer – foreground inactivity lock
- FLAG_SECURE in ALL activities (BaseActivity + ChatMediaActivity, SettingsActivity, PairingActivity)
- Key fingerprint verification (KeyFingerprintActivity)
- Auth sign-in flow (SignInActivity -> PairingActivity -> ConversationListActivity)
- FCM token registration via FcmTokenHelper.register() after sign-in, UID persisted to duoshield_prefs

Messaging

- Text, image, voice note messages (MessageBuilder)
- Delivery receipts ✓✓ (DeliveryReceiptHelper)
- Read receipts – teal ticks (ReadReceiptHelper + MessageStatusHelper)
- Message status ticks: pending=single grey, sent/delivered=double grey, read=double teal
- Message editing – 48h window (EditMessageHelper), (edited) label displayed in adapter
- Message deletion (own messages, soft-delete)
- Forwarding (ForwardMessageHelper)
- Emoji reactions (ReactMessageHelper + EmojiHelper)
- Message pinning – max 3 (PinMessageHelper)
- Reply with preview (swipe gesture via SwipeToReplyCallback)
- Disappearing messages (Off/1min/5min/1hr/1day, SelfDestructScheduler)
- Self-destruct countdown view (SelfDestructCountdownView)
- Self-destruct dual-sweep: deleteExpired(now) + deleteOlderThan(cutoff) in SelfDestructWorker
- Message search – offline Room (SearchHelper + MessageSearchActivity)
- Inline markdown (*bold* _italic_ ~strike~ `mono`) via TextStyleHelper
- Link URL extraction + preview chip (LinkPreviewHelper)
- Scroll-to-bottom FAB + unread counter badge
- Group message multi-select mode (MessageAdapter.enterMultiSelect/exitMultiSelect/toggleSelected/getSelectedIds)
- Bulk copy/delete/forward from ChatMediaActivity action dialog
- Export chat to PDF (ExportHelper – two-column layout, timestamps, edited badges, FileProvider share)

Presence

- Typing indicator (Firestore + TypingThrottle 2s debounce)
- Online dot (Firestore + OnlinePresenceHelper set/clear)

- 'last seen X min ago' – LastSeenFormatter
- Animated typing dots (TypingDotsAnimator)

List / Navigation

- Conversation list with DiffUtil (ConversationAdapter)
- LiveData conversation list (ConversationViewModel)
- LiveData message list (MessageViewModel)
- Swipe-to-delete with Snackbar undo (SwipeToDeleteCallback)
- In-list search filtering (no extra reads)

Media

- Camera + gallery picker (MediaPickerHelper)
- Image compression before upload (UploadHelper)
- Voice recording + waveform (VoiceRecorderHelper + VoiceWaveformView)
- Full-screen image viewer – FullScreenImageActivity (ScaleGestureDetector, FLAG_SECURE, registered in manifest)
- Voice note playback – VoiceNotePlayer (MediaPlayer wrapper, play/pause/resume/stop, released in onDestroy)
- Secure share via FileProvider (SecureShareHelper)
- Disk image cache 50 MB (ImageCacheHelper)
- Standardized Glide loading (GlideHelper)
- Storage usage estimate (MediaSizeEstimator)

Notifications

- FCM rich notifications (MessagingStyle + BigTextStyle for long bodies)
- Two channels: messages + silent
- setCategory(CATEGORY_MESSAGE)
- Inline reply from tray (MessageReplyReceiver)
- Mark as Read from tray (MarkReadReceiver)
- Auto-cancel on chat open

Background

- Self-destruct WorkManager jobs (SelfDestructScheduler)
- Weekly cache cleanup (StorageCleanupWorker)
- Backward message pagination – 30/page (MessagePaginator)
- Batch delivery receipts (WriteBatch, up to 500 per commit)

Infrastructure

- DuoShieldApp.java – Application subclass, calls NotificationStyler.createChannels(this), implements Configuration.Provider for WorkManager
- FileProvider registered in AndroidManifest for SecureShareHelper + ExportHelper (res/xml/file_paths.xml: cache-path, files-path, external-files-path)
- ACCESS_NETWORK_STATE permission added to manifest
- SearchView in toolbar -> SearchHelper.runSearch() / clearSearch() -> re-attaches live listener

UI Polish

- Slide-in/out activity transitions (4 directions)
- Message bubble pop-in animation
- Reaction emoji bounce animation
- Wrong PIN shake + haptic (HapticHelper)
- Send/mic button morph (SendButtonAnimator)
- Date separator chips (DateHeaderHelper)
- Offline banner (NetworkStateHelper)
- Firebase quota card (FirebaseQuotaSummary + layout_quota_stats.xml)

Remaining / Future Work

The following items are **NOT YET IMPLEMENTED**. Build these next (in order).

A	ECDH key exchange in PairingActivity	Wire ECDHHelper + PairingManager into PairingActivity so that key exchange happens at pairing time and the shared secret is stored securely. KeyFingerprintActivity already shows the fingerprint but the actual ECDH handshake needs to be triggered here.
---	--------------------------------------	---

B	Read receipts badge in ConversationListActivity	Show unread badge in ConversationsActivity conversation rows when partner has not yet read the last message. Tie into ReadReceiptHelper.
C	Badge count on launcher icon	Use ShortcutBadger library or the native NotificationManager badge API to reflect total unread count on the app icon.
D	Block screenshot in app switcher (verify FLAG_SECURE)	FLAG_SECURE is already set in BaseActivity. Verify it also applies to the app switcher thumbnail on a physical device. If not, override getWindow() flags in DuoShieldApp or each activity onCreate.

Strict Rules for Agent

1. DO NOT re-implement any feature listed in 'Fully Implemented Features'
2. DO NOT break MessageAdapter constructor signature
3. NO Cloud Functions – client-side + WorkManager only
4. Room DB = local source of truth – Firestore for live updates only
5. One Firestore listener per screen – detach in onStop()
6. Next Room DB version = 8, migration = MIGRATION_7_8
7. Always call FirebaseCostGuard before any Firestore read/write/delete
8. Batch deletes – use WorkManager or WriteBatch, never one-by-one
9. Return only new/modified files – never rewrite untouched files
10. FLAG_SECURE in ALL activities – already in BaseActivity, do not remove
11. Never use Unicode superscript/subscript in any generated output (use XML tags in ReportLab)
12. Migration chain for AppDatabase.java: 1→2→3→4→5→6→7 all registered. Next = MIGRATION_7_8

How to Continue in a New Replit Session

- Open a new Replit Agent session
- Paste THIS ENTIRE DOCUMENT as the first message
- Tell the agent what to build from 'Remaining / Future Work' above
- Reference duoshield_ui/ for all existing files – the agent can read them
- The agent will NOT repeat any existing work

Config / Rules Files

```
duoshield_ui/
■■■ firestore.rules ← Firestore security rules
■■■ storage.rules ← Firebase Storage security rules
■■■ app_dependencies.gradle ← All Gradle deps to add to app/build.gradle
■■■ AndroidManifest_additions.txt ← Activities + permissions to add to manifest
■ (includes FileProvider + ACCESS_NETWORK_STATE)
■■■ HOW_TO_INTEGRATE.md ← Step-by-step Android Studio wiring guide
■■■ res/xml/network_security_config.xml ← Network security config (no cleartext)
■■■ res/xml/file_paths.xml ← FileProvider paths (cache, files, external)
```

Deploy Firestore rules with: `firebase deploy --only firestore:rules`

Key Firestore security rule principles:

- Read/write only for conversation members
- Message create only if sender == auth.uid
- Message update only if sender == auth.uid (or reaction field only)
- Presence fields only writable if field name ends in own UID
- No client-side deletes of conversation documents (use hidden_ soft delete)